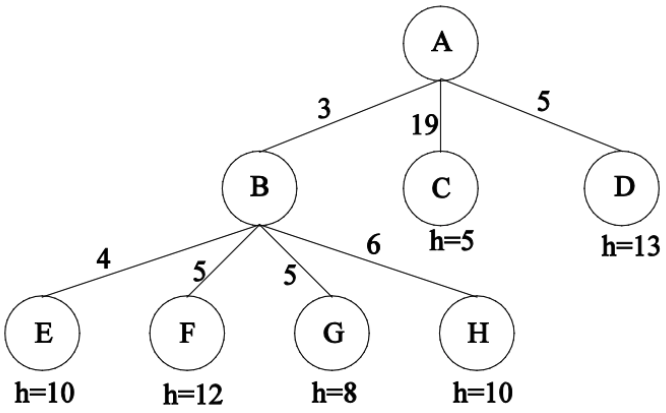
Lahore University of Management Sciences
Syed Babar Ali School of Science and Engineering
AI-500 – Fall 2025 – Midterm Exam – Part II

Campus ID: _____ **Signature:** _____

Note: Attempt all the questions (total 50 points)
Time allowed: 2 hours

Do Not Open This Exam Unless Told To Do So

Problem 1. Consider the following search tree produced after expanding nodes A and B, where each arc is labeled with the cost of the corresponding operator, and the leaves are labeled with the value of a heuristic function, h. For uninformed searches, assume children are expanded left to right. In case of ties, expand in alphabetical order.



Which one node will be expanded next by each of the following search methods?

(a) (**3 pts**) Depth-First search.
E

(b) (**3 pts**) Greedy Best-First search.
C (smallest h value)

(c) (**3 pts**) Uniform-Cost search.
D (smallest g value)

(d) (**3 pts**) A* search.
G (smallest g+h value)

Problem 2. Consider the following game tree. The root is a maximizing node, and children are visited left to right.

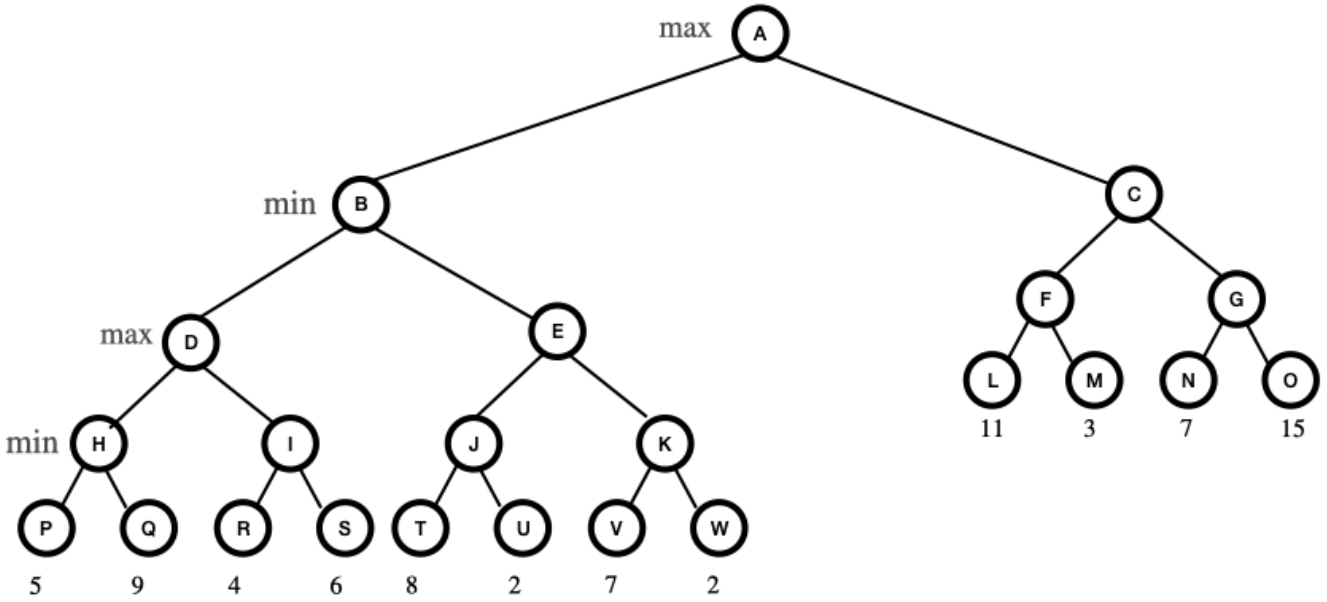


Figure 1: Alpha-Beta Pruning.

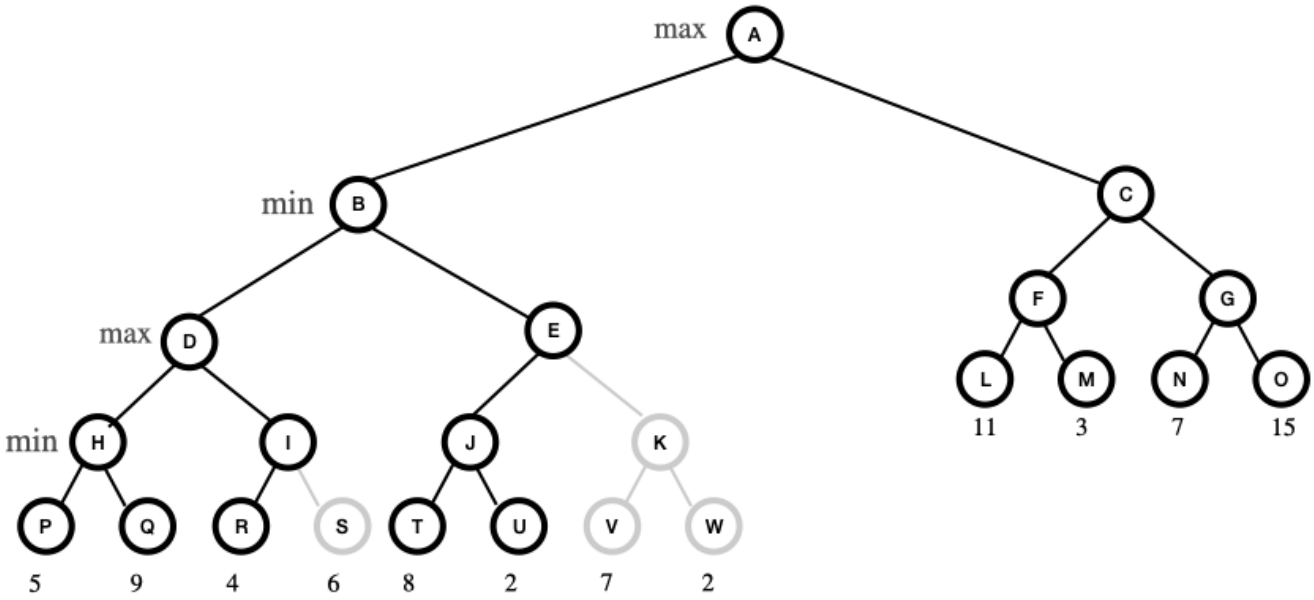


Figure 2: Alpha-Beta Pruning.

Using Alpha/Beta pruning answer the following:

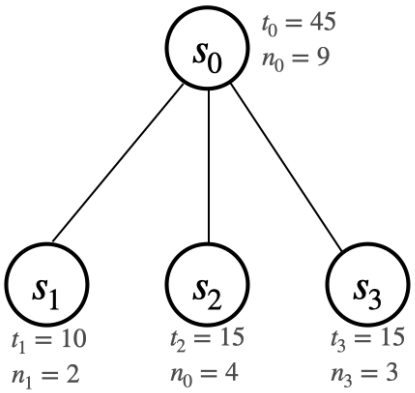
(a) (**2 pts**) Pruning occurs at the arc from I to S. [True/False]. True

(b) (**2 pts**) Pruning occurs at the arc from J to U. [True/False] False

(c) (**2 pts**) Pruning occurs at the arc from E to K. [True/False]
True

(d) (**2 pts**) Pruning occurs at the arc from G to H. [True/False]
False

Problem 3. Given the following tree, perform the selection step of Monte Carlo Tree Search (MCTS) (t_i = cumulative reward, n_i = number of times node s_i is visited).



(a) (**4 pts**) Using exploration parameter $c = 0.25$.

$$UCT(s_1) = 10/2 + 0.25\sqrt{ln(9)/2} = 5.2620$$
$$UCT(s_2) = 15/4 + 0.25\sqrt{ln(9)/4} = 3.9353$$
$$UCT(s_3) = 15/3 + 0.25\sqrt{ln(9)/3} = 5.2140$$

Hence node s_1 will be selected.

(b) (**4 pts**) Using exploration parameter $c = 3$.

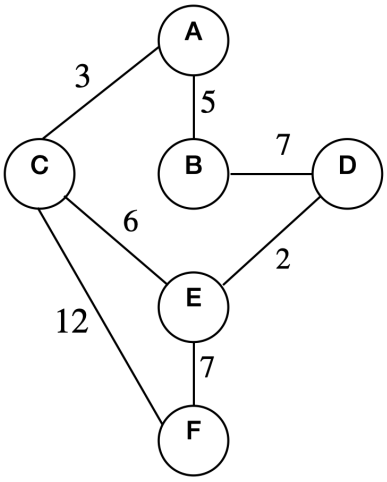
$$UCT(s_1) = 10/2 + 3\sqrt{ln(9)/2} = 8.1444$$
$$UCT(s_2) = 15/4 + 3\sqrt{ln(9)/4} = 5.9735$$
$$UCT(s_3) = 15/3 + 3\sqrt{ln(9)/3} = 7.5674$$

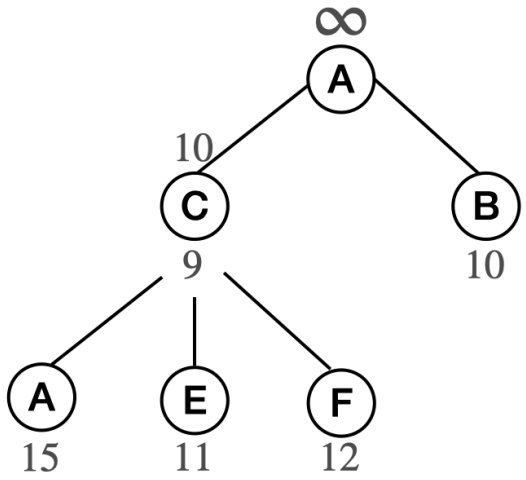
Hence node s_1 will be selected.

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Problem 4. (**10 pts**) Given the following graph, perform the Recursive Best First Search till the return of first failure).

A	9
B	5
C	5
D	7
E	4
F	0





A.f = 0+9 = 9

B.f = max(B.g+B.h,A.f)

B.f = max(5+5,9) = 10

C.f = max(C.g+C.h,A.f)

C.f = max(3+5,9) = 9

Best = C

$C.f > f_{limit}$

9 ¿ ∞

alt = B

$f_{limit} = min(f_{limit}, alt.f)$

$f_{limit} = min(\infty, 10) = 10$

A.f = max(3+3+9,9) = 15

E.f = max(7+4,9) = 11

F.f = max(12 + 0,9) = 12

Best = E

$E.f > f_{limit}$

11 > 10 return failure

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Problem 5. (**12 pts**) We want to build a decision tree using the following dataset.

ID	StudyHours	Attendance	Sleep	Pass
1	High	High	Adequate	Yes
2	High	Medium	Adequate	Yes
3	Medium	High	Less	Yes
4	Medium	Low	Adequate	No
5	Low	High	Adequate	Yes
6	Low	Low	Less	No
7	High	Low	Less	No
8	Low	Medium	Less	No

Using information gain find the best attribute for the root node.

Information Gain Calculation for Decision Tree

Step 2: Entropy of the Target Variable

The target attribute is **Pass** with two classes: Yes and No.

$$p_{Yes} = \frac{4}{8} = 0.5, \quad p_{No} = \frac{4}{8} = 0.5$$

Hence the entropy of the target variable is:

$$H(\text{Pass}) = -\sum_i p_i \log_2 p_i = -[0.5 \log_2 0.5 + 0.5 \log_2 0.5] = 1.0$$

$$\boxed{H(\text{Pass}) = 1.0 \text{ bit}}$$

—

Step 3: Information Gain for Each Attribute

(a) StudyHours Possible values: High, Medium, Low.

StudyHours	Yes	No	Total	Entropy
High	2	1	3	$-[(\frac{2}{3}) \log_2 \frac{2}{3} + (\frac{1}{3}) \log_2 \frac{1}{3}] = 0.918$
Medium	1	1	2	1.000
Low	1	2	3	0.918

Weighted remainder:

$$\begin{aligned} R(\text{StudyHours}) &= \frac{3}{8}(0.918) + \frac{2}{8}(1.000) + \frac{3}{8}(0.918) \\ &= 0.9386 \end{aligned}$$

Information gain:

$$IG(\text{StudyHours}) = 1.0 - 0.9386 = 0.0614$$

—

(b) Attendance Possible values: High, Medium, Low.

Attendance	Yes	No	Total	Entropy
High	3	0	3	0.000
Medium	1	1	2	1.000
Low	0	3	3	0.000

Weighted remainder:

$$R(\text{Attendance}) = \frac{3}{8}(0.0) + \frac{2}{8}(1.0) + \frac{3}{8}(0.0) = 0.25$$

Information gain:

$$IG(\text{Attendance}) = 1.0 - 0.25 = 0.75$$

—

(c) Sleep Possible values: Adequate, Less.

Sleep	Yes	No	Total	Entropy
Adequate	3	1	4	$-[(\frac{3}{4})\log_2 \frac{3}{4} + (\frac{1}{4})\log_2 \frac{1}{4}] = 0.811$
Less	1	3	4	0.811

Weighted remainder:

$$R(\text{Sleep}) = \frac{4}{8}(0.811) + \frac{4}{8}(0.811) = 0.811$$

Information gain:

$$IG(\text{Sleep}) = 1.0 - 0.811 = 0.189$$

—

Step 4: Selecting the Root Attribute

Attribute	Information Gain
StudyHours	0.061
Attendance	**0.750**
Sleep	0.189

The attribute with the highest information gain is:

$$\boxed{\text{Root Attribute} = \text{Attendance}}$$

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